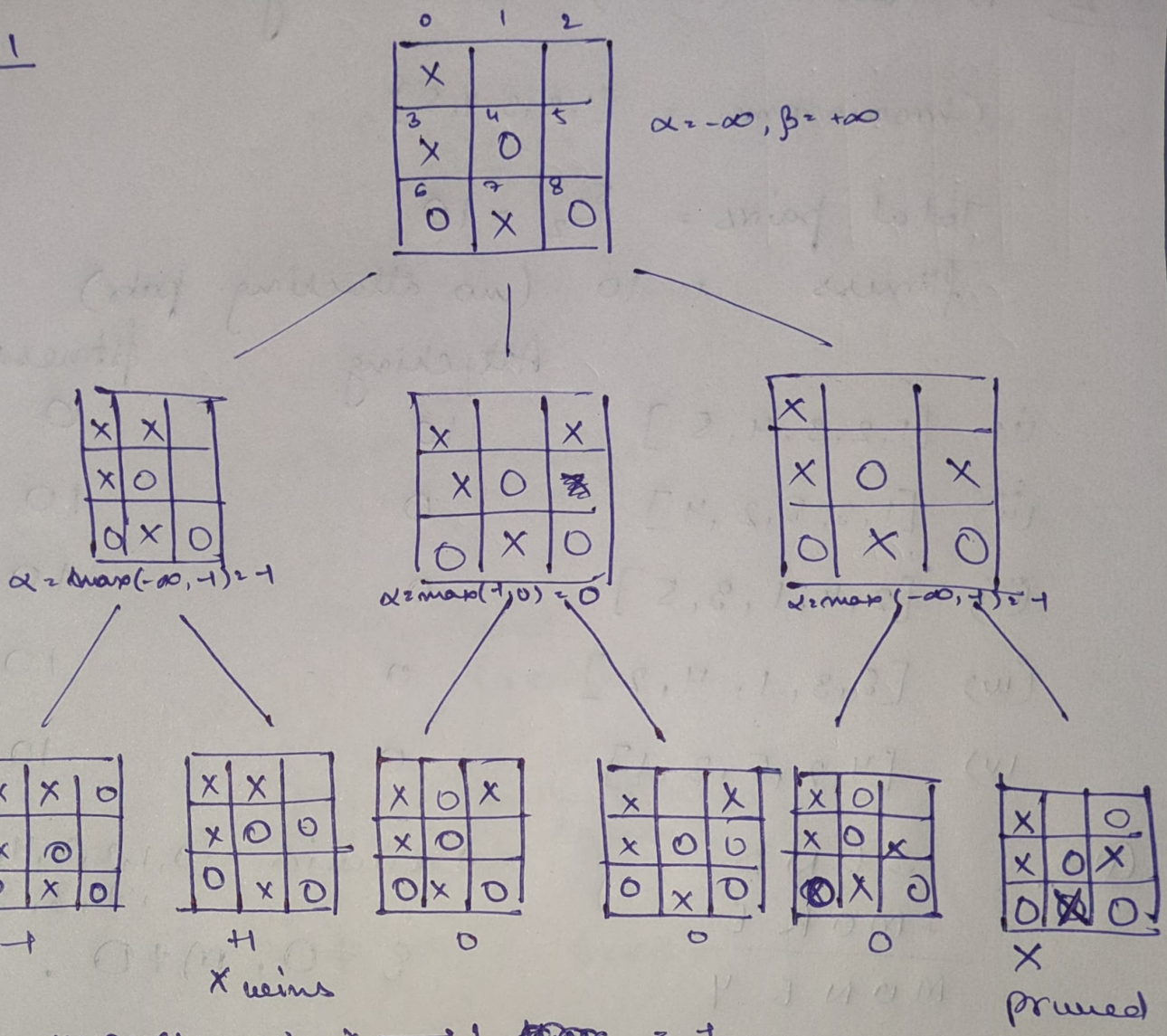
FOAI — Assignment - 5

Q.1



moves → X → 1 , = -1

X → 2 = 0

X → 5 = -1

X → 2

final solⁿ.

or

draw.

Q.2 N-Queens (5×5) Genetic Algorithm

Chromosome = [1, 2, 3, 4, 5]

Total pairs = ${}^5C_2 = 10$

fitness = 10 (no attacking pair)

		Attacking	fitness
(i)	[1, 2, 3, 4, 5]	10	0
(ii)	[1, 3, 5, 2, 4]	0	10
(iii)	[2, 4, 1, 3, 5]	0	10
(iv)	[5, 3, 1, 4, 2]	0	10
(v)	[4, 2, 5, 3, 1]	0	10

Q.3

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  SEND
 +MORE
 -----
 MONEY
```

Domain = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}

$S \neq 0$, $M \neq 0$.

$D + E = Y + 10$

$N + R = E + 10$

$E + O = N + 10$

$S + M = 0$

Start

M = 1

S = 9

O = 0

N = E + 1

R = 8

E = 5

N = 6

Y = 2

D = 7

Solution →

S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
Y = 2
R = 8

Q.5 Graph Coloring (Australian Map).

Colors
{red, green, blue}

Western Australia (red)

Northern Territory (green)

South Australia (blue)

Queensland (red)

New South Wales (green)

Victoria (red)

Tasmania (red).